Discrete Mathematics and Graph Theory

Sets and Functions

Department of Electronics and Telecommunication Engineering

Ref: Kenneth Rosen, Discrete Mathematics and its Applications, 7th Edition, McGraw Hill Publications, 2012

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Sets and Functions

Module 2

- Size of a set, Cartesian products, Venn diagrams, Set operations, Set identities, Cardinality of sets
- One-to-One and Onto Functions, Inverse Functions and Compositions of Functions
- Relations and their properties, Functions as Relations, Relations on a set, Representing relations, Closures of relations, Warshall‟s algorithm, Equivalence relations, Recurrence relations

Sets

- Sets are used to group objects together. Often, but not always, the objects in a set have similar properties. For instance, all the students who are currently enrolled in a school make up a set.
- A set is an unordered collection of objects, called elements or members of the set. A set is said to contain its elements.
- We write $a \in A$ to denote that a is an element of the set A. The notation $a \notin A$ denotes that a is not an element of the set A.
- The notation {a, b, c, d} represents the set with the four elements a, b, c, and d. This way of describing a set is known as the roster method.
- **EXAMPLE 1** The set V of all vowels in the English alphabet can be written as V = {a, e, i, o, u}.
- **EXAMPLE 2** The set O of odd positive integers less than 10 can be expressed by O = {1, 3, 5, 7, 9}.

Set

Set Builder notation

- We characterize all those elements in the set by stating the property or properties they must have to be members.
- For instance, the set O of all odd positive integers less than 10 can be written as

O = {x | x is an odd positive integer less than 10},or, specifying the universe as the set of positive integers, as O = {x ∈ Z+ | x is odd and x < 10}.

The following sets,, play an important role in discrete mathematics:

N = {0, 1, 2, 3, . . .}, the set of natural numbers
Z = {. . . ,−2,−1, 0, 1, 2, . . .}, the set of integers
Z+ = {1, 2, 3, . . .}, the set of positive integers
Q = {p/q | p ∈ Z, q ∈ Z, and q = 0}, the set of rational numbers
R, the set of real numbers
R+, the set of positive real numbers
C, the set of complex numbers.

Set of Real Numbers

When a and b are real numbers with
$a < b$, we write
$[a, b] = \{x \mid a \leq x \leq b\}$
$[a, b) = \{x \mid a \leq x < b\}$
$(a, b] = \{x \mid a < x \leq b\}$
$(a, b) = \{x \mid a < x < b\}$
Note that $[a, b]$ is called the **closed interval** from a to b and (a, b) is called the **open interval** from a to b.

Two sets are *equal* if and only if they have the same elements. Therefore, if A and B are sets, then A and B are equal if and only if $\forall x(x \in A \leftrightarrow x \in B)$. We write $A = B$ if A and B are equal sets.

Sets

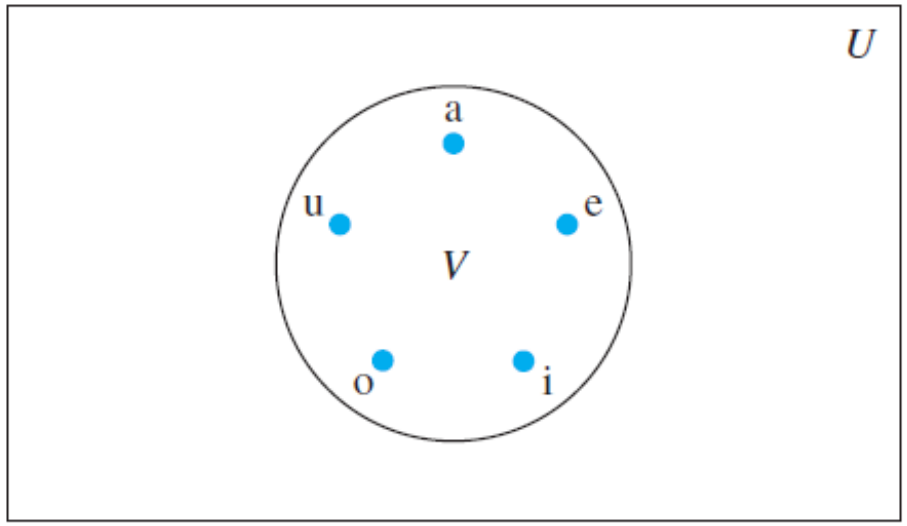
- The sets {1, 3, 5} and {3, 5, 1} are equal, because they have the same elements. Note that the order in which the elements of a set are listed does not matter.
- It does not matter if an element of a set is listed more than once, so {1, 3, 3, 3, 5, 5, 5, 5} is the same as the set {1, 3, 5} because they have the same elements.

- THE EMPTY SET There is a special set that has no elements. This set is called the empty set, or null set, and is denoted by ∅. The empty set can also be denoted by { }

- A set of elements with certain properties turns out to be the null set.
- The set of all positive integers that are greater than their squares is the null set.
- A set with one element is called a singleton set.

Venn diagram

Venn diagram: We draw a rectangle to indicate the universal set U, which is the set of the 26 letters of the English alphabet. Inside this rectangle we draw a circle to represent V . Inside this circle we indicate the elements of V with points



Subsets

The set A is a subset of B if and only if every element of A is also an element of B. We use the notation $A \subseteq B$ to indicate that A is a subset of the set B.

We see that $A \subseteq B$ if and only if the quantification $\forall x(x \in A \rightarrow x \in B)$ is true.
To show that A is not a subset of B we need only find one element $x \in A$ with $x \notin B$.

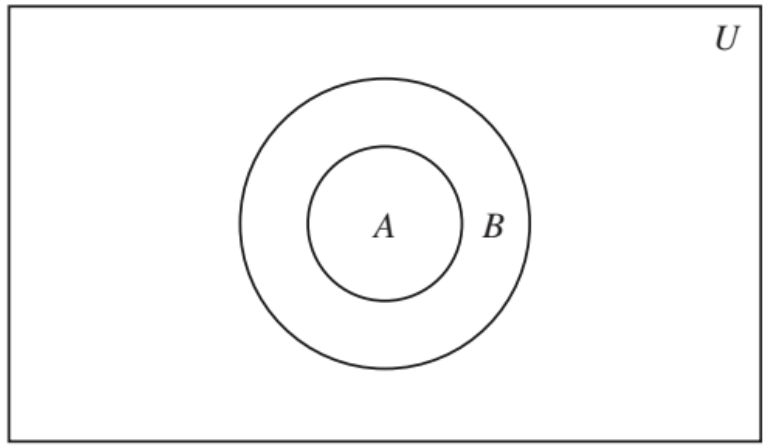
Showing that A is a Subset of B To show that $A \subseteq B$, show that if x belongs to A then x also belongs to B.

Showing that A is Not a Subset of B To show that $A \not\subseteq B$, find a single $x \in A$ such that $x \notin B$.

The set of all odd positive integers less than 10 is a subset of the set of all positive integers less than 10, the set of rational numbers is a subset of the set of real numbers,

The set of integers with squares less than 100 is not a subset of the set of nonnegative integers because -1 is in the former set [as $(-1)_2 < 100$], but not the later set.

Subset, Proper Set and Equal sets



Venn Diagram Showing that A Is a Subset of B.

For every set S, $(i)\ \emptyset \subseteq S$ and $(ii)\ S \subseteq S$.

A set A is a subset of a set B but that A ≠ B, we write A ⊂ B and say that A is a proper subset of B. For A ⊂ B to be true, it must be the case that A ⊆ B and there must exist an element x of B that is not an element of A. That is, A is a proper subset of B if and only if ∀x(x ∈ A → x ∈ B) ∧ ∃x(x ∈ B ∧ x ∉ A) is true.

Showing Two Sets are Equal To show that two sets A and B are equal, show that $A \subseteq B$ and $B \subseteq A$.

Size of a Set

Let S be a set. If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is a *finite set* and that n is the *cardinality* of S. The cardinality of S is denoted by $|S|$.

Let A be the set of odd positive integers less than 10. Then $|A| = 5$.

Let S be the set of letters in the English alphabet. Then $|S| = 26$.

Because the null set has no elements, it follows that $|\emptyset| = 0$.

Power Sets

Given a set S, the *power set* of S is the set of all subsets of the set S. The power set of S is denoted by $\mathcal{P}(S)$.

What is the power set of the set {0, 1, 2}?
Solution: The power set P({0, 1, 2}) is the set of all subsets of {0, 1, 2}. Hence, P({0, 1, 2}) = {∅, {0}, {1}, {2}, {0, 1}, {0, 2}, {1, 2}, {0, 1, 2}}.

Note that the empty set and the set itself are members of this set of subsets.

What is the power set of the empty set? What is the power set of the set {∅}?
Solution: The empty set has exactly one subset, namely, itself. Consequently, P(∅) = {∅}.
The set {∅} has exactly two subsets, namely, ∅ and the set {∅} itself. Therefore, P({∅}) = {∅, {∅}}. ▲
Note If a set has n elements, then its power set has 2n elements.

Cartesian Products

Let A and B be sets. The *Cartesian product* of A and B, denoted by $A \times B$, is the set of all ordered pairs (a, b), where $a \in A$ and $b \in B$. Hence,

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}.$$

What is the Cartesian product of A = {1, 2} and B = {a, b, c}?

Solution: The Cartesian product A × B is A × B = {(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)}.

Note that the Cartesian products A × B and B × A are not equal, unless A = ∅ or B = ∅ (so that A × B = ∅) or A = B

Cartesian Products Examples

What is the Cartesian product $A \times B \times C$, where $A = \{0, 1\}$, $B = \{1, 2\}$, and $C = \{0, 1, 2\}$?

Solution: The Cartesian product $A \times B \times C$ consists of all ordered triples (a, b, c), where $a \in A$, $b \in B$, and $c \in C$. Hence,

$A \times B \times C = \{(0, 1, 0), (0, 1, 1), (0, 1, 2), (0, 2, 0), (0, 2, 1), (0, 2, 2), (1, 1, 0), (1, 1, 1), (1, 1, 2), (1, 2, 0), (1, 2, 1), (1, 2, 2)\}$. Remark: Note that when A, B, and C are sets, $(A \times B) \times C$ is not the same as $A \times B \times C$

We use the notation A^2 to denote $A \times A$, the Cartesian product of the set A with itself. Similarly, $A^3 = A \times A \times A$, $A^4 = A \times A \times A \times A$, and so on. More generally,

$A^n = \{(a_1, a_2, \ldots, a_n) \mid a_i \in A \text{ for } i = 1, 2, \ldots, n\}$.

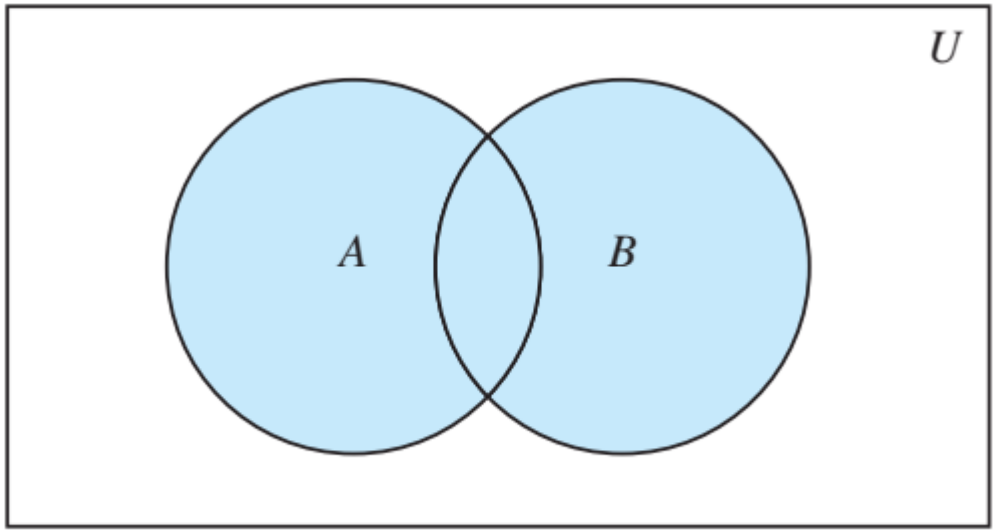
Cartesian Products Examples

- Suppose that A = {1, 2}. It follows that A^2 = {(1, 1), (1, 2), (2, 1), (2, 2)} and A^3 = {(1, 1, 1), (1, 1, 2), (1, 2, 1), (1, 2, 2), (2, 1, 1), (2, 1, 2), (2, 2, 1), (2, 2, 2)}.
- A subset R of the Cartesian product A × B is called a relation from the set A to the set B.
- The elements of R are ordered pairs, where the first element belongs to A and the second to B.
- For example, R = {(a, 0), (a, 1), (a, 3), (b, 1), (b, 2), (c, 0), (c, 3)} is a relation from the set {a, b, c} to the set {0, 1, 2, 3}.
- A relation from a set A to itself is called a relation on A.

Set Operations

Let A and B be sets. The *union* of the sets A and B, denoted by $A \cup B$, is the set that contains those elements that are either in A or in B, or in both.

$$A \cup B = \{x \mid x \in A \vee x \in B\}.$$



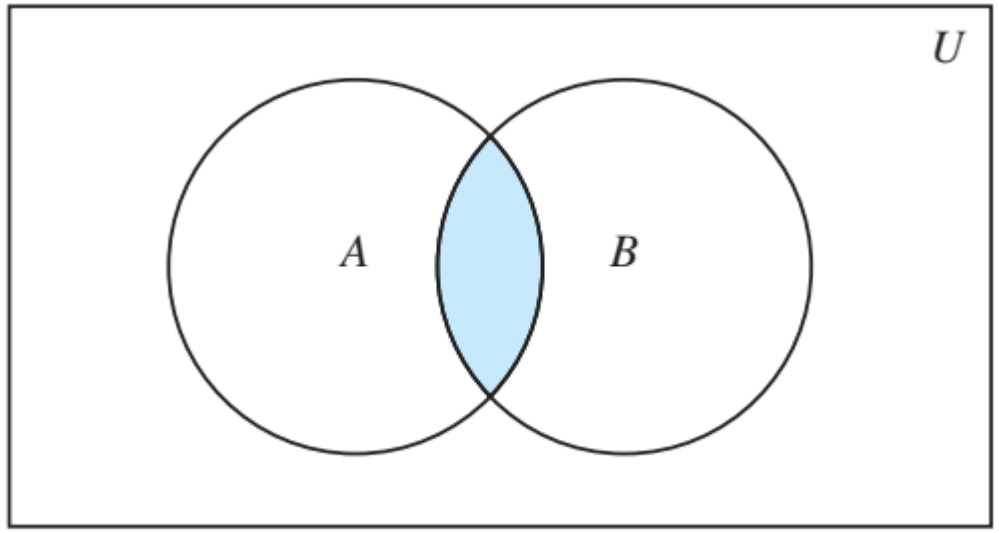
$A \cup B$ is shaded.

The union of the sets {1, 3, 5} and {1, 2, 3} is the set {1, 2, 3, 5}; that is, {1, 3, 5} ∪ {1, 2, 3} = {1, 2, 3, 5}.

Set Operations

Let A and B be sets. The *intersection* of the sets A and B, denoted by $A \cap B$, is the set containing those elements in both A and B.

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$



$A \cap B$ is shaded.

The intersection of the sets {1, 3, 5} and {1, 2, 3} is the set {1, 3}; that is,
{1, 3, 5} ∩ {1, 2, 3} = {1, 3}.

Disjoint Sets and Cardinality of Union

Two sets are called *disjoint* if their intersection is the empty set.

Let A = {1, 3, 5, 7, 9} and B = {2, 4, 6, 8, 10}. Because A ∩ B = ∅, A and B are disjoint.

Cardinality of Union

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

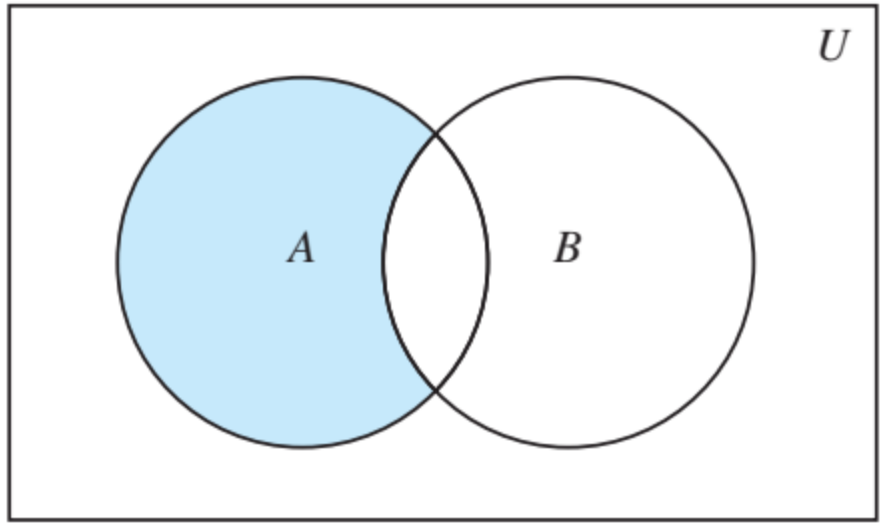
Differences of Sets

Let A and B be sets. The *difference* of A and B, denoted by $A - B$, is the set containing those elements that are in A but not in B. The difference of A and B is also called the *complement of B with respect to A*.

Remark: The difference of sets A and B is sometimes denoted by $A \setminus B$.

The difference of {1, 3, 5} and {1, 2, 3} is the set {5}; that is, {1, 3, 5} - {1, 2, 3} = {5}. This is different from the difference of {1, 2, 3} and {1, 3, 5}, which is the set {2}.

$$A - B = A \cap \overline{B}.$$

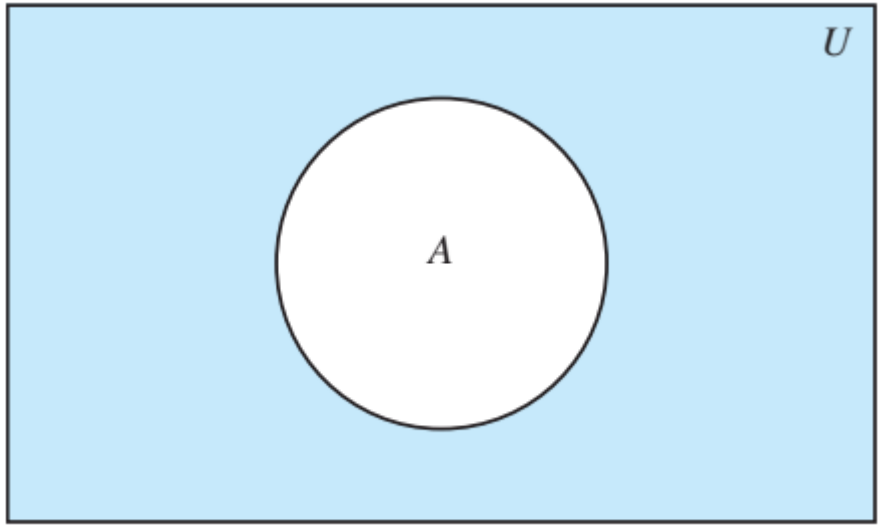


$A - B$ is shaded.

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Complement of a Set

Let U be the universal set. The *complement* of the set A, denoted by $\overline{A}$, is the complement of A with respect to U. Therefore, the complement of the set A is $U - A$.



$\overline{A}$ is shaded.

Let A = {a, e, i, o, u} (where the universal set is the set of letters of the English alphabet). Then
A = {b, c, d, f, g, h, j, k, l, m, n, p, q, r, s, t, v, w, x, y, z}.

Set Identities

TABLE 1 Set Identities.

Identity	*Name*
$A \cap U = A$ $A \cup \emptyset = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{(\overline{A})} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$	Associative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$\overline{A \cap B} = \overline{A} \cup \overline{B}$ $\overline{A \cup B} = \overline{A} \cap \overline{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \overline{A} = U$ $A \cap \overline{A} = \emptyset$	Complement laws

Examples on Set Identities

Use set builder notation and logical equivalences to establish the first De Morgan law

$$\overline{A \cap B} = \overline{A} \cup \overline{B}.$$

Solution: We can prove this identity with the following steps.

$$\begin{aligned}
\overline{A \cap B} &= \{x \mid x \notin A \cap B\} && \text{by definition of complement}\\
&= \{x \mid \neg(x \in (A \cap B))\} && \text{by definition of does not belong symbol}\\
&= \{x \mid \neg(x \in A \wedge x \in B)\} && \text{by definition of intersection}\\
&= \{x \mid \neg(x \in A) \vee \neg(x \in B)\} && \text{by the first De Morgan law for logical equivalences}\\
&= \{x \mid x \notin A \vee x \notin B\} && \text{by definition of does not belong symbol}\\
&= \{x \mid x \in \overline{A} \vee x \in \overline{B}\} && \text{by definition of complement}\\
&= \{x \mid x \in \overline{A} \cup \overline{B}\} && \text{by definition of union}\\
&= \overline{A} \cup \overline{B} && \text{by meaning of set builder notation}
\end{aligned}$$

Examples on Set Identities

Use a membership table to show that A ∩ (B ∪ C) = (A ∩ B) ∪ (A ∩ C).

TABLE 2 A Membership Table for the Distributive Property.

A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$A \cap C$	$(A \cap B) \cup (A \cap C)$
1	1	1	1	1	1	1	1
1	1	0	1	1	1	0	1
1	0	1	1	1	0	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

Examples on Set Identities

Let A, B, and C be sets. Show that

$$\overline{A \cup (B \cap C)} = (\overline{C} \cup \overline{B}) \cap \overline{A}.$$

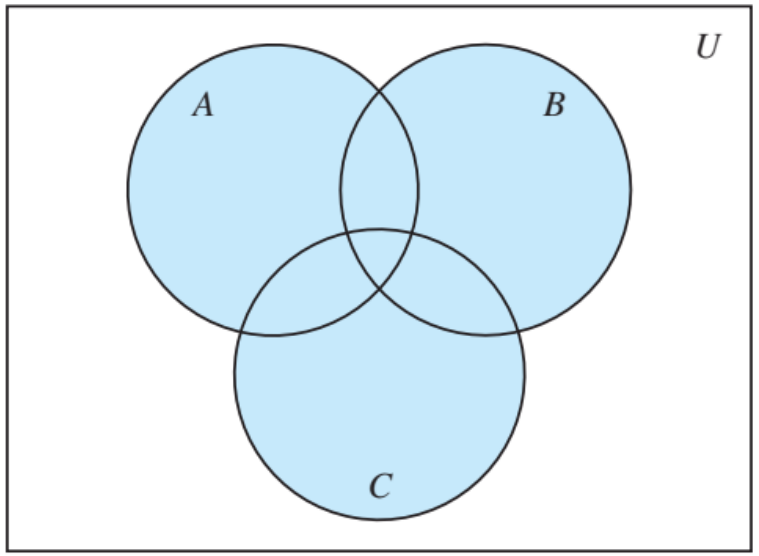
Solution: We have

$$\begin{aligned}
\overline{A \cup (B \cap C)} &= \overline{A} \cap \overline{(B \cap C)} && \text{by the first De Morgan law} \\
&= \overline{A} \cap (\overline{B} \cup \overline{C}) && \text{by the second De Morgan law} \\
&= (\overline{B} \cup \overline{C}) \cap \overline{A} && \text{by the commutative law for intersections} \\
&= (\overline{C} \cup \overline{B}) \cap \overline{A} && \text{by the commutative law for unions.}
\end{aligned}$$

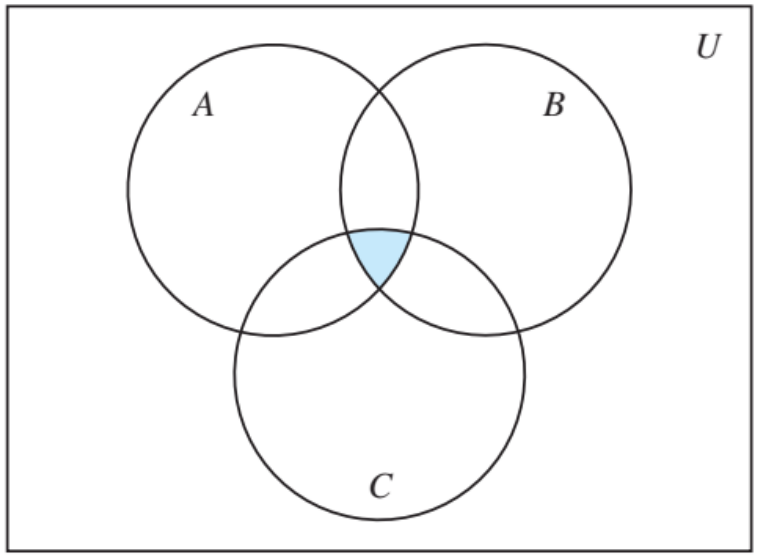
Generalized Unions and Intersections

The *union* of a collection of sets is the set that contains those elements that are members of at least one set in the collection.

The *intersection* of a collection of sets is the set that contains those elements that are members of all the sets in the collection.



(a) $A \cup B \cup C$ is shaded.



(b) $A \cap B \cap C$ is shaded.

Generalized Unions and Intersections

Let $A = \{0, 2, 4, 6, 8\}$, $B = \{0, 1, 2, 3, 4\}$, and $C = \{0, 3, 6, 9\}$. What are $A \cup B \cup C$ and $A \cap B \cap C$?

Solution: The set $A \cup B \cup C$ contains those elements in at least one of A, B, and C. Hence,

$$A \cup B \cup C = \{0, 1, 2, 3, 4, 6, 8, 9\}.$$

The set $A \cap B \cap C$ contains those elements in all three of A, B, and C. Thus,

$$A \cap B \cap C = \{0\}.$$

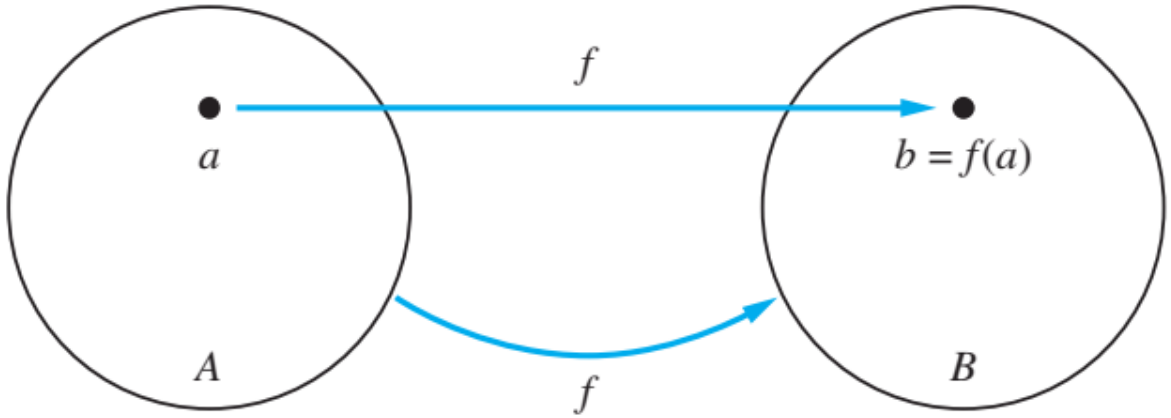
Functions

Let A and B be nonempty sets. A *function* f from A to B is an assignment of exactly one element of B to each element of A. We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A. If f is a function from A to B, we write $f : A \rightarrow B$.

If f is a function from A to B, we say that A is the *domain* of f and B is the *codomain* of f. If $f(a) = b$, we say that b is the *image* of a and a is a *preimage* of b. The *range*, or *image*, of f is the set of all images of elements of A. Also, if f is a function from A to B, we say that f *maps* A to B.

Two functions are equal when they have the same domain, have the same codomain, and map each element of their common domain to the same element in their common codomain.

Functions



The Function f Maps A to B.

Let f : Z → Z assign the square of an integer to this integer. Then, f (x) = x^2, where the domain of f is the set of all integers, the codomain of f is the set of all integers, and the range of f is the set of all integers that are perfect squares, namely, {0, 1, 4, 9, . . . }.

Functions

Let f_1 and f_2 be functions from A to $\mathbf{R}$. Then $f_1 + f_2$ and $f_1 f_2$ are also functions from A to $\mathbf{R}$ defined for all $x \in A$ by

$$(f_1 + f_2)(x) = f_1(x) + f_2(x), \qquad (f_1 f_2)(x) = f_1(x) f_2(x).$$

Let f_1 and f_2 be functions from $\mathbf{R}$ to $\mathbf{R}$ such that $f_1(x) = x^2$ and $f_2(x) = x - x^2$. What are the functions $f_1 + f_2$ and $f_1 f_2$?

Solution: From the definition of the sum and product of functions, it follows that

$$(f_1 + f_2)(x) = f_1(x) + f_2(x) = x^2 + (x - x^2) = x$$

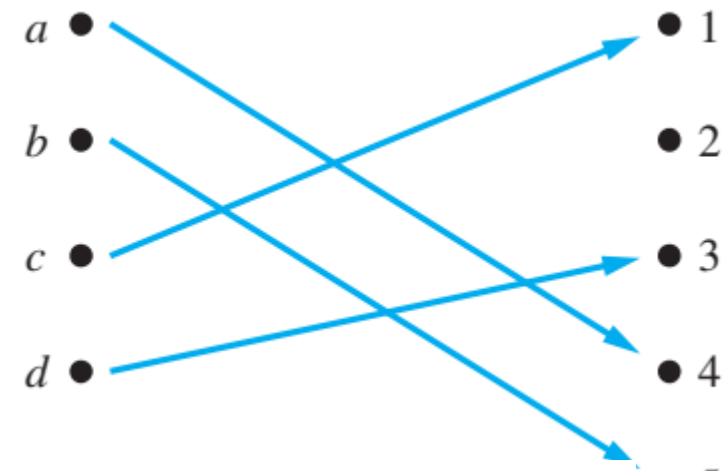
and

$$(f_1 f_2)(x) = x^2(x - x^2) = x^3 - x^4.$$

One-to-One Functions

A function f is said to be *one-to-one*, or an *injunction*, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f. A function is said to be *injective* if it is one-to-one.

$$\forall a \forall b (f(a) = f(b) \rightarrow a = b)$$



A One-to-One Function.

Determine whether the function f from {a, b, c, d} to {1, 2, 3, 4, 5} with f (a) = 4, f (b) = 5, f (c) = 1, and f (d) = 3 is one-to-one.

Solution: The function f is one-to-one because f takes on different values at the four elements of its domain.

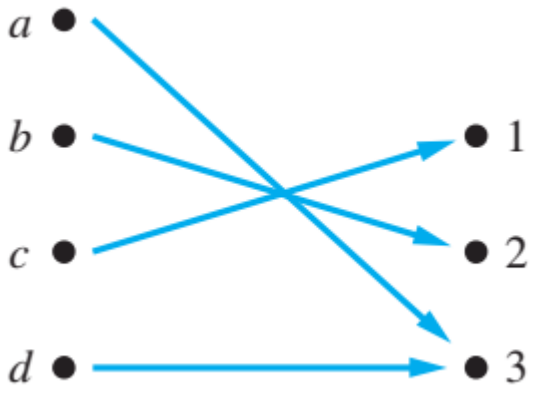
Determine whether the function f (x) = x^2 from the set of integers to the set of integers is one-to-one.

Solution: The function f (x) = x^2 is not one-to-one because, for instance, f (1) = f (-1) = 1, but 1 = -1.

Onto Functions

A function f from A to B is called *onto*, or a *surjection*, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$. A function f is called *surjective* if it is onto.

$$\forall y \exists x (f(x) = y)$$



An Onto Function.

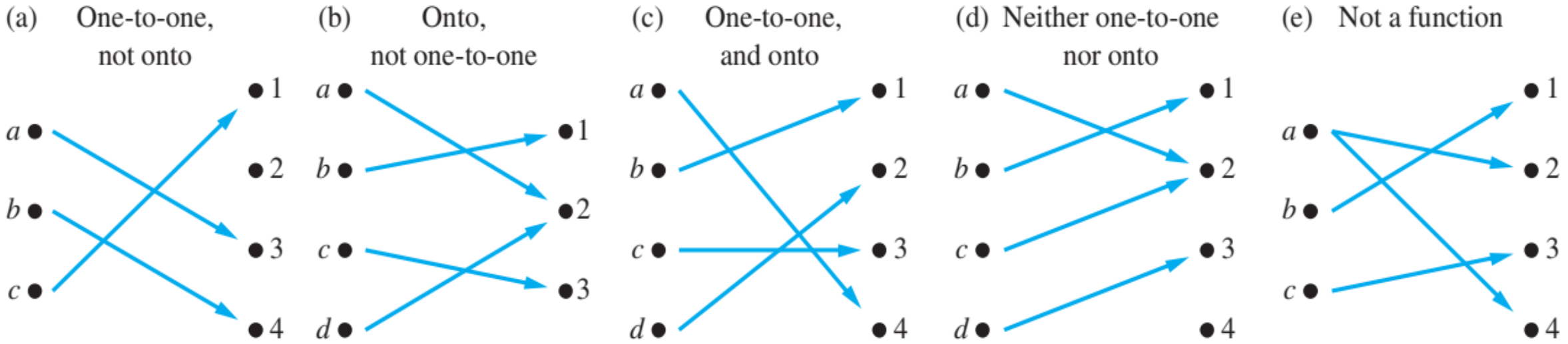
Let f be the function from {a, b, c, d} to {1, 2, 3} defined by f (a) = 3, f (b) = 2, f (c) = 1, and f (d) = 3. Is f an onto function?

Solution: Because all three elements of the codomain are images of elements in the domain, we see that f is onto. This is illustrated in Figure 4. Note that if the codomain were {1, 2, 3, 4}, then f would not be onto.

Is the function f (x) = x^2 from the set of integers to the set of integers onto?

Solution: The function f is not onto because there is no integer x with x^2 = -1, for instance.

Examples of Different Types of Correspondences



Bijection and Identity Function

The function f is a *one-to-one correspondence*, or a *bijection*, if it is both one-to-one and onto. We also say that such a function is *bijective*.

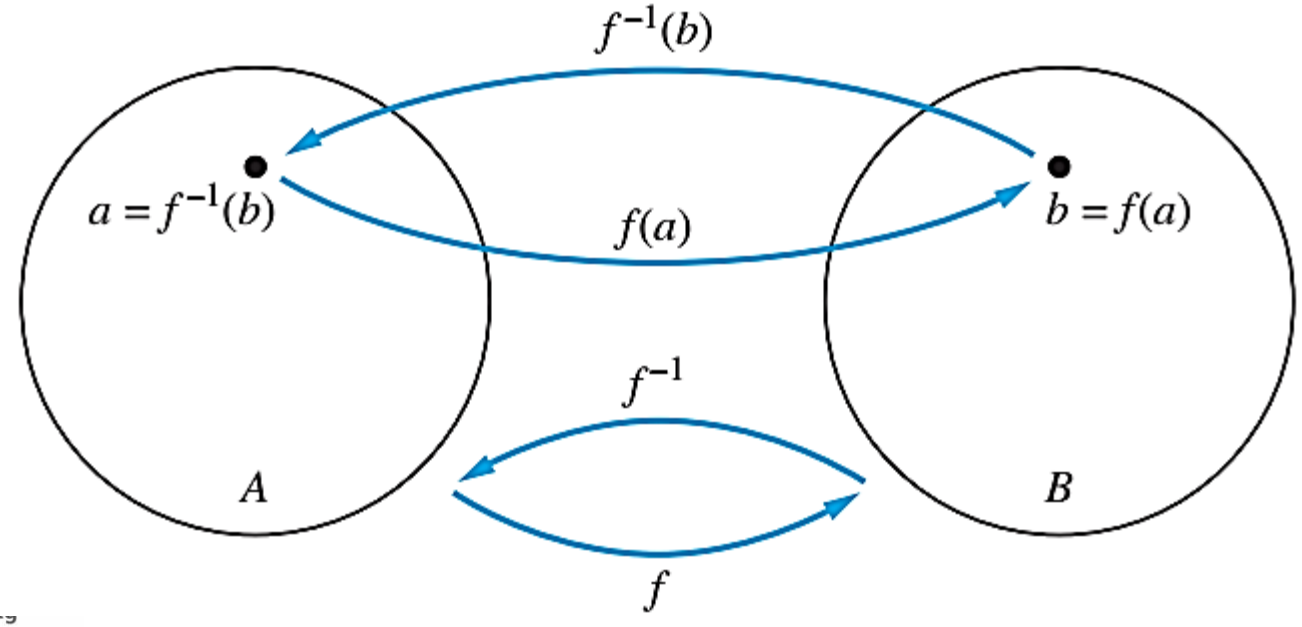
Let f be the function from {a, b, c, d} to {1, 2, 3, 4} with f (a) = 4, f (b) = 2, f (c) = 1, and f (d) = 3. Is f a bijection?

Solution: The function f is one-to-one and onto. It is one-to-one because no two values in the domain are assigned the same function value. It is onto because all four elements of the codomain are images of elements in the domain. Hence, f is a bijection.

Let A be a set. The identity function on A is the function $\iota A : A \rightarrow A$, where $\iota A(x) = x$

Inverse Functions

Let f be a one-to-one correspondence from the set A to the set B. The *inverse function* of f is the function that assigns to an element b belonging to B the unique element a in A such that $f(a) = b$. The inverse function of f is denoted by f^{-1}. Hence, $f^{-1}(b) = a$ when $f(a) = b$.



Invertible and Non Invertible Functions

A one-to-one correspondence is called invertible because we can define an inverse of this function. A function is not invertible if it is not a one-to-one correspondence, because the inverse of such a function does not exist.

Let f be the function from $\{a, b, c\}$ to $\{1, 2, 3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Is f invertible, and if it is, what is its inverse?

Solution: The function f is invertible because it is a one-to-one correspondence. The inverse function f^{-1} reverses the correspondence given by f, so $f^{-1}(1) = c$, $f^{-1}(2) = a$, and $f^{-1}(3) = b$.

Invertible and Non Invertible Functions Examples

Let $f : \mathbf{Z} \to \mathbf{Z}$ be such that $f(x) = x + 1$. Is f invertible, and if it is, what is its inverse?

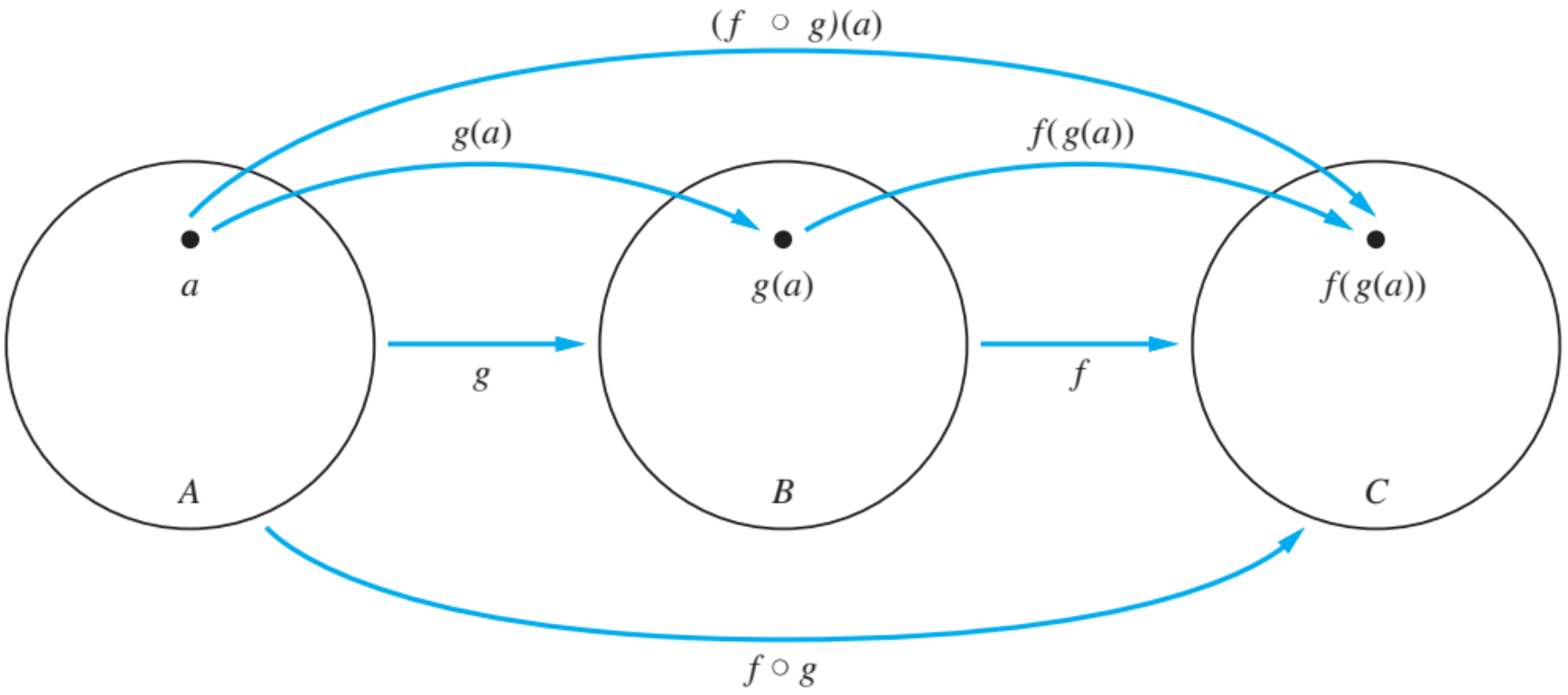
Solution: The function f has an inverse because it is a one-to-one correspondence, as follows from Examples 10 and 14. To reverse the correspondence, suppose that y is the image of x, so that $y = x + 1$. Then $x = y - 1$. This means that $y - 1$ is the unique element of $\mathbf{Z}$ that is sent to y by f. Consequently, $f^{-1}(y) = y - 1$.

Let f be the function from $\mathbf{R}$ to $\mathbf{R}$ with $f(x) = x^2$. Is f invertible?

Solution: Because $f(-2) = f(2) = 4$, f is not one-to-one. If an inverse function were defined, it would have to assign two elements to 4. Hence, f is not invertible. (Note we can also show that f is not invertible because it is not onto.)

Compositions of Functions

Let g be a function from the set A to the set B and let f be a function from the set B to the set C. The *composition* of the functions f and g, denoted for all $a \in A$ by $f \circ g$, is defined by $(f \circ g)(a) = f(g(a))$.



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Compositions of Functions Examples

Let g be the function from the set $\{a, b, c\}$ to itself such that $g(a) = b$, $g(b) = c$, and $g(c) = a$. Let f be the function from the set $\{a, b, c\}$ to the set $\{1, 2, 3\}$ such that $f(a) = 3$, $f(b) = 2$, and $f(c) = 1$. What is the composition of f and g, and what is the composition of g and f?

Solution: The composition $f \circ g$ is defined by $(f \circ g)(a) = f(g(a)) = f(b) = 2$, $(f \circ g)(b) = f(g(b)) = f(c) = 1$, and $(f \circ g)(c) = f(g(c)) = f(a) = 3$.

Note that $g \circ f$ is not defined, because the range of f is not a subset of the domain of g. ◀

Compositions of Functions Examples

Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g? What is the composition of g and f?

Solution: Both the compositions $f \circ g$ and $g \circ f$ are defined. Moreover,

$$(f \circ g)(x) = f(g(x)) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 7$$

and

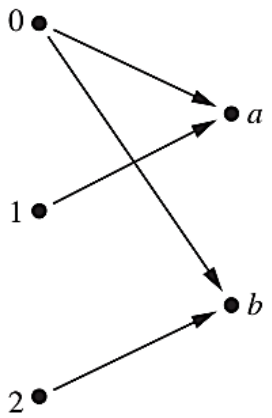
$$(g \circ f)(x) = g(f(x)) = g(2x + 3) = 3(2x + 3) + 2 = 6x + 11.$$

Relations

Relationships between elements of sets are represented using the structure called a relation, which is just a subset of the Cartesian product of the sets.

Let A and B be sets. A *binary relation from* A *to* B is a subset of $A \times B$.

Let A = {0, 1, 2} and B = {a, b}. Then {(0, a), (0, b), (1, a), (2, b)} is a relation from A to B. This means, for instance, that 0 R a, but that 1R b. Relations can be represented graphically, as shown in Figure, using arrows to represent ordered pairs. Another way to represent this relation is to use a table, which is also done in Figure



R	a	b
0	×	×
1	×	
2		×

Functions as Relations

The graph of f is the set of ordered pairs (a, b) such that b = f (a). Because the graph of f is a subset of A × B, it is a relation from A to B.

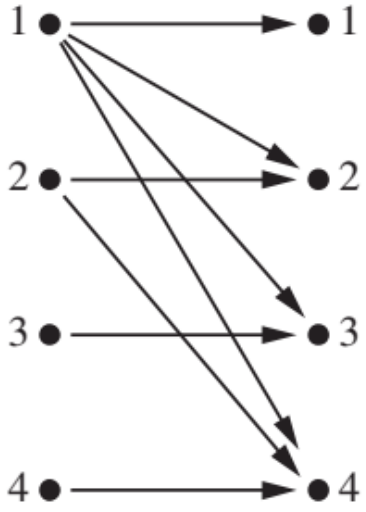
Conversely, if R is a relation from A to B such that every element in A is the first element of exactly one ordered pair of R, then a function can be defined with R as its graph.

This can be done by assigning to an element a of A the unique element b ∈ B such that (a, b) ∈ R.

Relations on a Set

A *relation on a set* A is a relation from A to A.

Let A be the set {1, 2, 3, 4}. Which ordered pairs are in the relation R = {(a, b) | a divides b}?
Solution: Because (a, b) is in R if and only if a and b are positive integers not exceeding 4 such that a divides b, we see that R = {(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)}.



R	1	2	3	4
1	×	×	×	×
2		×		×
3			×	
4				×

Relations on a Set Examples

Consider these relations on the set of integers:
R1 = {(a, b) | a ≤ b},
R2 = {(a, b) | a > b},
R3 = {(a, b) | a = b or a = -b},
R4 = {(a, b) | a = b},
R5 = {(a, b) | a = b + 1},
R6 = {(a, b) | a + b ≤ 3}
Which of these relations contain each of the pairs (1, 1), (1, 2), (2, 1), (1, −1), and (2, 2)?

The pair (1, 1) is in R1, R3, R4, and R6; (1, 2) is in R1 and R6; (2, 1) is in R2, R5, and R6; (1, -1) is in R2, R3, and R6; and finally, (2, 2) is in R1, R3, and R4.

Relations on a Set Examples

How many relations are there on a set with n elements?

Solution: A relation on a set A is a subset of $A \times A$. Because $A \times A$ has n^2 elements when A has n elements, and a set with m elements has 2^m subsets, there are 2^{n^2} subsets of $A \times A$. Thus, there are 2^{n^2} relations on a set with n elements. For example, there are $2^{3^2} = 2^9 = 512$ relations on the set $\{a, b, c\}$.

Properties of Relations

A relation R on a set A is called *reflexive* if $(a, a) \in R$ for every element $a \in A$.

Consider the following relations on {1, 2, 3, 4}:
R1 = {(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)}, R2 = {(1, 1), (1, 2), (2, 1)},
R3 = {(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)}, R4 = {(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)},
R5 = {(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)}, R6 = {(3, 4)}.
Which of these relations are reflexive?

Solution: The relations R3 and R5 are reflexive because they both contain all pairs of the form (a, a), namely, (1, 1), (2, 2), (3, 3), and (4, 4). The other relations are not reflexive because they do not contain all of these ordered pairs. In particular, R1, R2, R4, and R6 are not reflexive because (3, 3) is not in any of these relations.

Properties of Relations

A relation R on a set A is called *symmetric* if $(b, a) \in R$ whenever $(a, b) \in R$, for all $a, b \in A$. A relation R on a set A such that for all $a, b \in A$, if $(a, b) \in R$ and $(b, a) \in R$, then $a = b$ is called *antisymmetric*.

- That is, a relation is symmetric if and only if a is related to b implies that b is related to a.
- A relation is antisymmetric if and only if there are no pairs of distinct elements a and b with a related to b and b related to a.
- That is, the only way to have a related to b and b related to a is for a and b to be the same element.
- The terms symmetric and antisymmetric are not opposites, because a relation can have both of these properties or may lack both of them

Remark: Using quantifiers, we see that the relation R on the set A is symmetric if $\forall a \forall b((a, b) \in R \rightarrow (b, a) \in R)$. Similarly, the relation R on the set A is antisymmetric if $\forall a \forall b(((a, b) \in R \wedge (b, a) \in R) \rightarrow (a = b))$.

Properties of Relations

Consider the following relations on {1, 2, 3, 4}:
R1 = {(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)}, R2 = {(1, 1), (1, 2), (2, 1)},
R3 = {(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)}, R4 = {(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)},
R5 = {(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)}, R6 = {(3, 4)}.
Which of these relations are symmetric and which are antisymmetric?

Solution: The relations R2 and R3 are symmetric, because in each case (b, a) belongs to the relation whenever (a, b) does. For R2, the only thing to check is that both (2, 1) and (1, 2) are in the relation. For R3, it is necessary to check that both (1, 2) and (2, 1) belong to the relation, and (1, 4) and (4, 1) belong to the relation\. None of the other relations is symmetric. This is done by finding a pair (a, b) such that it is in the relation but (b, a) is not.
R4, R5, and R6 are all antisymmetric. For each of these relations there is no pair of elements a and b with a = b such that both (a, b) and (b, a) belong to the relation. None of the other relations is antisymmetric. This is done by finding a pair (a, b) with a = b such that (a, b) and (b, a) are both in the relation.

Properties of Relations

Is the “divides” relation on the set of positive integers symmetric? Is it antisymmetric?

Solution: This relation is not symmetric because 1|2, but 2 | 1. It is antisymmetric, for if a and b are positive integers with a |b and b |a, then a = b

Properties of Relations

A relation R on a set A is called *transitive* if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$, for all $a, b, c \in A$.

Remark: Using quantifiers we see that the relation R on a set A is transitive if we have $\forall a \forall b \forall c(((a, b) \in R \wedge (b, c) \in R) \rightarrow (a, c) \in R)$.

Properties of Relations

Consider the following relations on {1, 2, 3, 4}:
R1 = {(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)}, R2 = {(1, 1), (1, 2), (2, 1)},
R3 = {(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)}, R4 = {(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)},
R5 = {(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)}, R6 = {(3, 4)}.
Which of these relations are are transitive?

Solution: R4, R5, and R6 are transitive. For each of these relations, we can show that it is transitive by verifying that if (a, b) and (b, c) belong to this relation, then (a, c) also does.
For instance, R4 is transitive, because (3, 2) and (2, 1), (4, 2) and (2, 1), (4, 3) and (3, 1), and (4, 3) and (3, 2) are the only such sets of pairs, and (3, 1), (4, 1), and (4, 2) belong to R4.
R5 and R6 are transitive.
R1 is not transitive because (3, 4) and (4, 1) belong to R1, but (3, 1) does not.
R2 is not transitive because (2, 1) and (1, 2) belong to R2, but (2, 2) does not.
R3 is not transitive because (4, 1) and (1, 2) belong to R3, but (4, 2) does not.

Properties of Relations

Is the “divides” relation on the set of positive integers transitive?

Solution: Suppose that a divides b and b divides c. Then there are positive integers k and l such that b = ak and c = bl. Hence, c = a(kl), so a divides c. It follows that this relation is transitive

Combining Relations

Let A = {1, 2, 3} and B = {1, 2, 3, 4}. The relations R1 = {(1, 1), (2, 2), (3, 3)} and R2 = {(1, 1), (1, 2), (1, 3), (1, 4)} can be combined to obtain

$R1 \cup R2 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (3, 3)\}$,
$R1 \cap R2 = \{(1, 1)\}$,
$R1 - R2 = \{(2, 2), (3, 3)\}$,
$R2 - R1 = \{(1, 2), (1, 3), (1, 4)\}$.

Combining Relations

Let A and B be the set of all students and the set of all courses at a school, respectively. Suppose that R1 consists of all ordered pairs (a, b), where a is a student who has taken course b, and R2 consists of all ordered pairs (a, b), where a is a student who requires course b to graduate. What are the relations R1 ∪ R2, R1 ∩ R2, R1 ⊕ R2, R1 - R2, and R2 - R1?

Solution: The relation R1 ∪ R2 consists of all ordered pairs (a, b), where a is a student who either has taken course b or needs course b to graduate, and R1 ∩ R2 is the set of all ordered pairs (a, b), where a is a student who has taken course b and needs this course to graduate.
Also, R1 ⊕ R2 consists of all ordered pairs (a, b), where student a has taken course b but does not need it to graduate or needs course b to graduate but has not taken it.
R1 - R2 is the set of ordered pairs (a, b), where a has taken course b but does not need it to graduate; that is, b is an elective course that a has taken.
R2 - R1 is the set of all ordered pairs (a, b), where b is a course that a needs to graduate but has not taken.

Composite Relations

Let R be a relation from a set A to a set B and S a relation from B to a set C. The *composite* of R and S is the relation consisting of ordered pairs (a, c), where $a \in A$, $c \in C$, and for which there exists an element $b \in B$ such that $(a, b) \in R$ and $(b, c) \in S$. We denote the composite of R and S by $S \circ R$.

What is the composite of the relations R and S, where R is the relation from{1, 2, 3} to{1, 2, 3, 4} with R = {(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)} and S is the relation from {1, 2, 3, 4} to {0, 1, 2} with S = {(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)}?

Solution: S ◦R is constructed using all ordered pairs in R and ordered pairs in S, where the second element of the ordered pair in R agrees with the first element of the ordered pair in S.

For example, the ordered pairs (2, 3) in R and (3, 1) in S produce the ordered pair (2, 1) in S ◦R.

Computing all the ordered pairs in the composite, we find S ◦ R = {(1, 0), (1, 1), (2, 1), (2, 2), (3, 0), (3, 1)}.

Representing Relations

There are many ways to represent a relation between finite sets

- List its ordered pairs.
- Use a table
- Uses zero–one matrices.
- Use pictorial representations called directed graphs

Representing Relations Using Matrices

Suppose that $A = \{1, 2, 3\}$ and $B = \{1, 2\}$. Let R be the relation from A to B containing (a, b) if $a \in A$, $b \in B$, and $a > b$. What is the matrix representing R if $a_1 = 1$, $a_2 = 2$, and $a_3 = 3$, and $b_1 = 1$ and $b_2 = 2$?

Solution: Because $R = \{(2, 1), (3, 1), (3, 2)\}$, the matrix for R is

$$\mathbf{M}_R = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

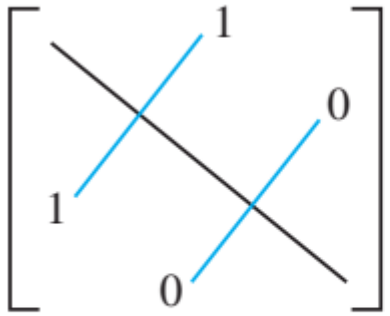
Representing Relations

Let $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3, b_4, b_5\}$. Which ordered pairs are in the relation R represented by the matrix

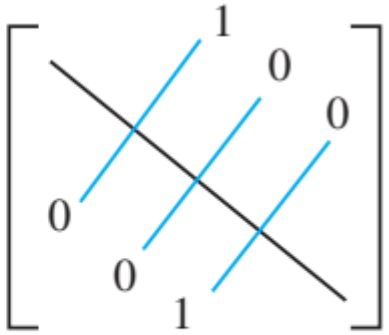
$$\mathbf{M}_R = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix}?$$

Solution: Because R consists of those ordered pairs (a_i, b_j) with $m_{ij} = 1$, it follows that R = {(a1, b2), (a2, b1), (a2, b3), (a2, b4), (a3, b1), (a3, b3), (a3, b5)}.

Zero–One Matrices for Symmetric and Antisymmetric Relations



(a) Symmetric



(b) Antisymmetric

Suppose that the relation R on a set is represented by the matrix

$$\mathbf{M}_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Is R reflexive, symmetric, and/or antisymmetric?

Solution: Because all the diagonal elements of this matrix are equal to 1, R is reflexive. Moreover, because MR is symmetric, it follows that R is symmetric. It is also easy to see that R is not antisymmetric.

Representing Relations

Suppose that the relations R_1 and R_2 on a set A are represented by the matrices

$$\mathbf{M}_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

What are the matrices representing $R_1 \cup R_2$ and $R_1 \cap R_2$?

Solution: The matrices of these relations are

$$\mathbf{M}_{R_1 \cup R_2} = \mathbf{M}_{R_1} \vee \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \quad \mathbf{M}_{R_1 \cap R_2} = \mathbf{M}_{R_1} \wedge \mathbf{M}_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Representing Relations

Find the matrix representing the relations $S \circ R$, where the matrices representing R and S are

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

Solution: The matrix for $S \circ R$ is

$$\mathbf{M}_{S \circ R} = \mathbf{M}_R \odot \mathbf{M}_S = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Representing Relations

Find the matrix representing the relation R^2, where the matrix representing R is

$$\mathbf{M}_R = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

Solution: The matrix for R^2 is

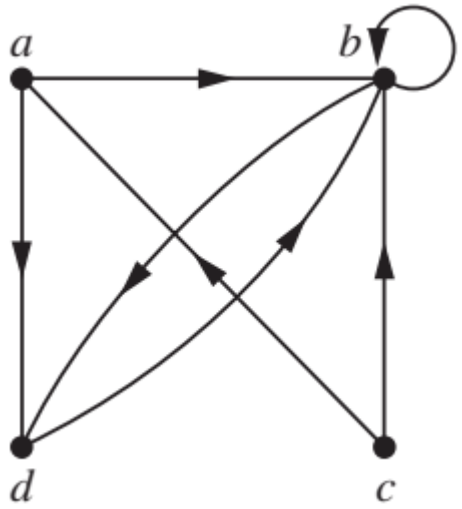
$$\mathbf{M}_{R^2} = \mathbf{M}_R^{[2]} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Representing Relations Using Digraphs

A *directed graph*, or *digraph*, consists of a set V of *vertices* (or *nodes*) together with a set E of ordered pairs of elements of V called *edges* (or *arcs*). The vertex a is called the *initial vertex* of the edge (a, b), and the vertex b is called the *terminal vertex* of this edge.

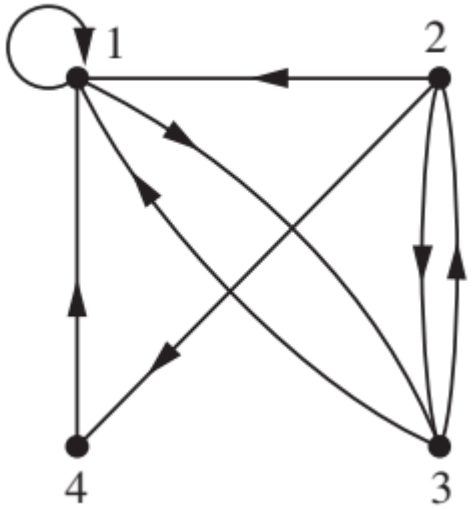
An edge of the form (a, a) is represented using an arc from the vertex a back to itself. Such an edge is called a loop.

The directed graph with vertices a, b, c, and d, and edges (a, b), (a, d), (b, b), (b, d), (c, a), (c, b), and (d, b) is displayed in Figure 3



Digraphs

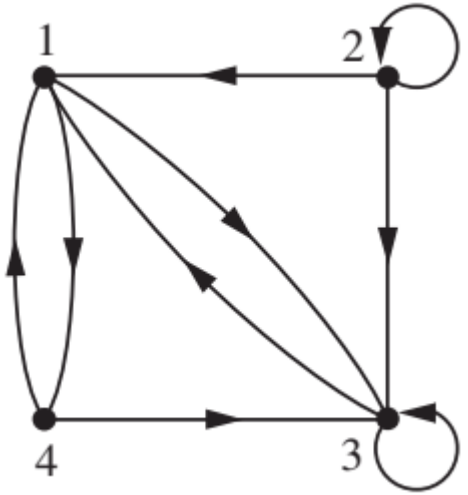
The directed graph of the relation R = {(1, 1), (1, 3), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (4, 1)} on the set {1, 2, 3, 4} is shown in Figure



What are the ordered pairs in the relation R represented by the directed graph shown in Figure 5?

Solution: The ordered pairs (x, y) in the relation are R = {(1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (3, 1), (3, 3), (4, 1), (4, 3)}.

Each of these pairs corresponds to an edge of the directed graph, with (2, 2) and (3, 3) corresponding to loops.



Closures of Relations

- In general, let R be a relation on a set A. R may or may not have some property P, such as reflexivity, symmetry, or transitivity.
- If there is a relation S with property P containing R such that S is a subset of every relation with property P containing R, then S is called the closure
- Types of closures
  - Reflexive
  - Symmetric
  - Transitive

Reflexive Closures

- The relation R = {(1, 1), (1, 2), (2, 1), (3, 2)} on the set A = {1, 2, 3} is not reflexive.
- We can produce a reflexive relation containing R that is as small as possible.
- This can be done by adding (2, 2) and (3, 3) to R, because these are the only pairs of the form (a, a) that are not in R. This new relation contains R.
- Furthermore, any reflexive relation that contains R must also contain (2, 2) and (3, 3).
- Because this relation contains R, is reflexive, and is contained within every reflexive relation that contains R, it is called the reflexive closure of R

What is the reflexive closure of the relation R = {(a, b) | a < b} on the set of integers?

Solution: The reflexive closure of R is

$R \cup \Delta = \{(a, b) \mid a < b\} \cup \{(a, a) \mid a \in Z\} = \{(a, b) \mid a \leq b\}$.

Symmetric Closure

- The relation {(1, 1), (1, 2), (2, 2), (2, 3), (3, 1), (3, 2)} on {1, 2, 3} is not symmetric.
- We can produce a symmetric relation that is as small as possible and contains R.
- We need only add (2, 1) and (1, 3), because these are the only pairs of the form (b, a) with (a, b) ∈ R that are not in R.
- This new relation is symmetric and contains R. Furthermore, any symmetric relation that contains R must contain this new relation, because a symmetric relation that contains R must contain (2, 1) and (1, 3).
- Consequently, this new relation is called the symmetric closure of R

What is the symmetric closure of the relation R = {(a, b) | a > b} on the set of positive integers?

Solution: The symmetric closure of R is the relation

$R \cup R^{-1} = \{(a, b) \mid a > b\} \cup \{(b, a) \mid a > b\} = \{(a, b) \mid a = b\}$.

This last equality follows because R contains all ordered pairs of positive integers where the first element is greater than the second element and R-1 contains all ordered pairs of positive integers where the first element is less than the second.

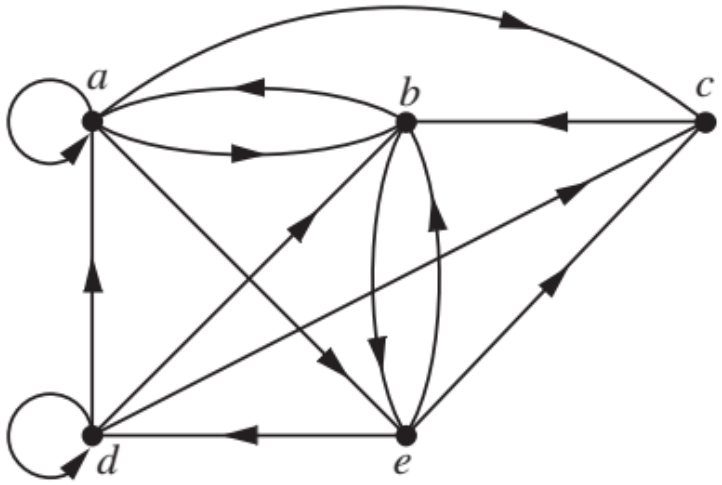
Paths in Directed Graphs

A path in a directed graph is obtained by traversing along edges (in the same direction as indicated by the arrow on the edge)

A *path* from a to b in the directed graph G is a sequence of edges (x_0, x_1), (x_1, x_2), $(x_2, x_3), \ldots, (x_{n-1}, x_n)$ in G, where n is a nonnegative integer, and $x_0 = a$ and $x_n = b$, that is, a sequence of edges where the terminal vertex of an edge is the same as the initial vertex in the next edge in the path. This path is denoted by $x_0, x_1, x_2, \ldots, x_{n-1}, x_n$ and has *length* n. We view the empty set of edges as a path of length zero from a to a. A path of length $n \geq 1$ that begins and ends at the same vertex is called a *circuit* or *cycle*.

A path in a directed graph can pass through a vertex more than once. Moreover, an edge in a directed graph can occur more than once in a path.

Paths in Directed Graphs



Which of the following are paths in the directed graph shown in Figure
: a, b, e, d; a, e, c, d, b;
b, a, c, b, a, a, b;
d, c; c, b, a; e, b, a, b, a, b, e?
What are the lengths of those that are paths?
Which of the paths in this list are circuits?

Solution:

- Because each of (a, b), (b, e), and (e, d) is an edge, a, b, e, d is a path of length three.
- Because (c, d) is not an edge, a, e, c, d, b is not a path.
- Also, b, a, c, b, a, a, b is a path of length six because (b, a), (a, c), (c, b), (b, a), (a, a), and (a, b) are all edges.
- We see that d, c is a path of length one, because (d, c) is an edge.
- Also c, b, a is a path of length two, because (c, b) and (b, a) are edges.
- All of (e, b), (b, a), (a, b), (b, a), (a, b), and (b, e) are edges, so e, b, a, b, a, b, e is a path of length six.
- The two paths b, a, c, b, a, a, b and e, b, a, b, a, b, e are circuits because they begin and
- end at the same vertex. The paths a, b, e, d; c, b, a; and d, c are not circuits.

Paths in Relations and Transitive Closures

Let R be a relation on a set A. There is a path of length n, where n is a positive integer, from a to b if and only if $(a, b) \in R^n$.

Let R be a relation on a set A. The *connectivity relation* R^* consists of the pairs (a, b) such that there is a path of length at least one from a to b in R.

Because R^n consists of the pairs (a, b) such that there is a path of length n from a to b, it follows that R^* is the union of all the sets R^n. In other words,

$$R^* = \bigcup_{n=1}^{\infty} R^n.$$

Transitive Closures

The transitive closure of a relation R equals the connectivity relation R^*.

Let A be a set with n elements, and let R be a relation on A. If there is a path of length at least one in R from a to b, then there is such a path with length not exceeding n. Moreover, when $a \neq b$, if there is a path of length at least one in R from a to b, then there is such a path with length not exceeding $n - 1$.

we see that the transitive closure of R is the union of R, R^2, R^3, . . . , and R^n. This follows because there is a path in R^* between two vertices if and only if there is a path between these vertices in R^i, for some positive integer i with $i \leq n$. Because

$$R^* = R \cup R^2 \cup R^3 \cup \cdots \cup R^n$$

Transitive Closures Examples

Find the zero–one matrix of the transitive closure of the relation R where

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Solution: By Theorem 3, it follows that the zero–one matrix of R^* is

$$\mathbf{M}_{R^*} = \mathbf{M}_R \vee \mathbf{M}_R^{[2]} \vee \mathbf{M}_R^{[3]}.$$

Because

$$\mathbf{M}_R^{[2]} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_R^{[3]} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix},$$

it follows that

$$\mathbf{M}_{R^*} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \vee \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

Warshall's Algorithm

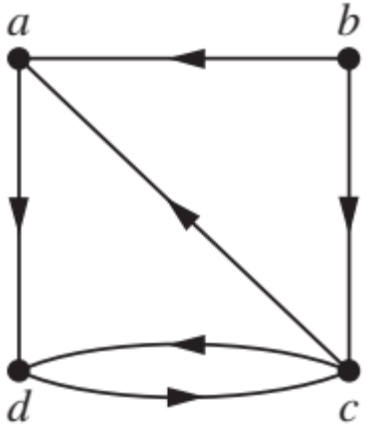
Warshall's algorithm is based on the construction of a sequence of zero–one matrices. These matrices are $\mathbf{W}_0, \mathbf{W}_1, \ldots, \mathbf{W}_n$, where $\mathbf{W}_0 = \mathbf{M}_R$ is the zero–one matrix of this relation, and $\mathbf{W}_k = [w_{ij}^{(k)}]$, where $w_{ij}^{(k)} = 1$ if there is a path from v_i to v_j such that all the interior vertices of this path are in the set $\{v_1, v_2, \ldots, v_k\}$ (the first k vertices in the list) and is 0 otherwise. (The first and last vertices in the path may be outside the set of the first k vertices in the list.) Note that $\mathbf{W}_n = \mathbf{M}_{R^*}$, because the (i, j)th entry of $\mathbf{M}_{R^*}$ is 1 if and only if there is a path from v_i to v_j, with all interior vertices in the set $\{v_1, v_2, \ldots, v_n\}$ (but these are the only vertices in the directed graph).

Warshall's Algorithm

ALGORITHM 2 Warshall Algorithm.

procedure *Warshall* ($\mathbf{M}_R$: $n \times n$ zero–one matrix)
$\mathbf{W} := \mathbf{M}_R$
for $k := 1$ **to** n
    for $i := 1$ **to** n
        for $j := 1$ **to** n
        $w_{ij} := w_{ij} \vee (w_{ik} \wedge w_{kj})$
return $\mathbf{W}\{\mathbf{W} = [w_{ij}]$ is $\mathbf{M}_{R^*}\}$

Warshall's Algorithm Example

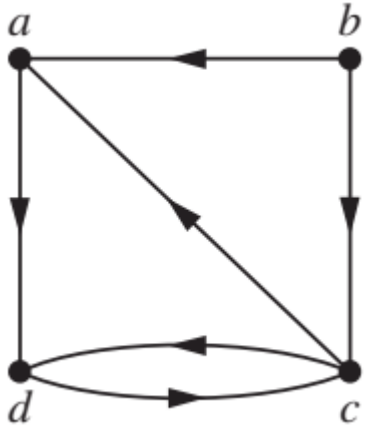


Let R be the relation with directed graph shown in Figure 3. Let a, b, c, d be a listing of the elements of the set. Find the matrices $\mathbf{W}_0$, $\mathbf{W}_1$, $\mathbf{W}_2$, $\mathbf{W}_3$, and $\mathbf{W}_4$. The matrix $\mathbf{W}_4$ is the transitive closure of R.

Solution: Let $v_1 = a$, $v_2 = b$, $v_3 = c$, and $v_4 = d$. $\mathbf{W}_0$ is the matrix of the relation. Hence,

$$\mathbf{W}_0 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

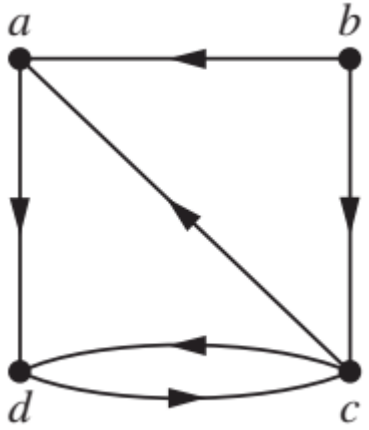
Warshall's Algorithm Example contd…



$\mathbf{W}_1$ has 1 as its (i, j)th entry if there is a path from v_i to v_j that has only $v_1 = a$ as an interior vertex. Note that all paths of length one can still be used because they have no interior vertices. Also, there is now an allowable path from b to d, namely, b, a, d. Hence,

$$\mathbf{W}_1 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

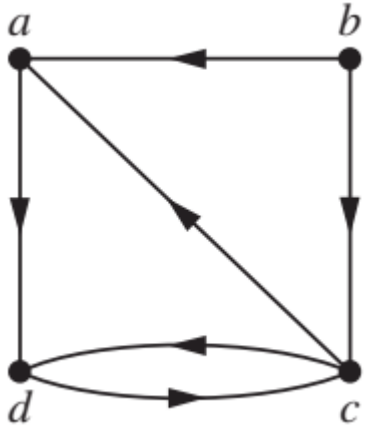
Warshall's Algorithm Example contd…



$\mathbf{W}_2$ has 1 as its (i, j)th entry if there is a path from v_i to v_j that has only $v_1 = a$ and/or $v_2 = b$ as its interior vertices, if any. Because there are no edges that have b as a terminal vertex, no new paths are obtained when we permit b to be an interior vertex. Hence, $\mathbf{W}_2 = \mathbf{W}_1$.

$$\mathbf{W}'_2 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

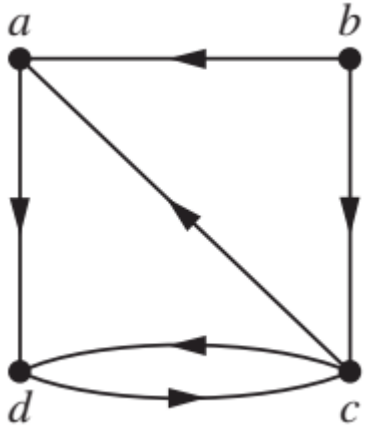
Warshall's Algorithm Example contd…



$\mathbf{W}_3$ has 1 as its (i, j)th entry if there is a path from v_i to v_j that has only $v_1 = a$, $v_2 = b$, and/or $v_3 = c$ as its interior vertices, if any. We now have paths from d to a, namely, d, c, a, and from d to d, namely, d, c, d. Hence,

$$\mathbf{W}_3 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}.$$

Warshall's Algorithm Example contd…



Finally, $\mathbf{W}_4$ has 1 as its (i, j)th entry if there is a path from v_i to v_j that has $v_1 = a$, $v_2 = b$, $v_3 = c$, and/or $v_4 = d$ as interior vertices, if any. Because these are all the vertices of the graph, this entry is 1 if and only if there is a path from v_i to v_j. Hence,

$$\mathbf{W}_4 = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}.$$

This last matrix, $\mathbf{W}_4$, is the matrix of the transitive closure.

Equivalence Relations

A relation on a set A is called an *equivalence relation* if it is reflexive, symmetric, and transitive.

Two elements a and b that are related by an equivalence relation are called *equivalent*. The notation $a \sim b$ is often used to denote that a and b are equivalent elements with respect to a particular equivalence relation.

Equivalence Relations Examples

Let R be the relation on the set of real numbers such that aRb if and only if a - b is an integer. Is R an equivalence relation?

Solution: Because a - a = 0 is an integer for all real numbers a,

- aRa for all real numbers a. Hence, R is reflexive.
- Now suppose that aRb. Then a - b is an integer, so b - a is also an integer.
- Hence, bRa. It follows that R is symmetric.
- If aRb and bRc, then a - b and b - c are integers.
- Therefore, a - c = (a - b) + (b - c) is also an integer. Hence, aRc.
- Thus, R is transitive. Consequently, R is an equivalence relation.

Equivalence Relations Examples

Let m be an integer with m > 1. Show that the relation R = {(a, b) | a ≡ b (mod m)} is an equivalence relation on the set of integers.

Solution: a ≡ b (mod m) if and only if m divides a - b.

- Note that a - a = 0 is divisible by m, because 0 = 0 · m. Hence, a ≡ a (mod m), so congruence modulo m is reflexive.
- Now suppose that a ≡ b (mod m). Then a - b is divisible by m, so a - b = km, where k is an integer. It follows that b - a = (-k)m, so b ≡ a (mod m). Hence, congruence modulo m is symmetric.
- Next, suppose that a ≡ b (mod m) and b ≡ c (mod m). Then m divides both a - b and b - c. Therefore, there are integers k and l with a - b = km and b - c = lm.
- Adding these two equations shows that a - c = (a - b) + (b - c) = km + lm = (k + l)m. Thus, a ≡ c (mod m). Therefore, congruence modulo m is transitive.
- It follows that congruence modulo m is an equivalence relation.

Equivalence Relations Examples

Show that the “divides” relation is the set of positive integers in not an equivalence relation

We know that the “divides” relation is reflexive and transitive.

We know that this relation is not symmetric (for instance, $2 \mid 4$ but $4 \nmid 2$). We conclude that the “divides” relation on the set of positive integers is not an equivalence relation.

Equivalence Relations Examples

Let R be the relation on the set of real numbers such that xR y if and only if x and y are real numbers that differ by less than 1, that is $|x - y| < 1$. Show that R is not an equivalence relation.

Solution: R is reflexive because $|x - x| = 0 < 1$
whenever $x \in R$. R is symmetric, for if xR y, where x and y are real numbers, then $|x - y| < 1$, which tells us that $|y - x| = |x - y| < 1$, so that yRx.
However, R is not an equivalence relation because it is not transitive.
Take $x = 2.8$, $y = 1.9$, and $z = 1.1$, so that $|x - y| = |2.8 - 1.9| = 0.9 < 1$, $|y - z| = |1.9 - 1.1| = 0.8 < 1$, but $|x - z| = |2.8 - 1.1| = 1.7 > 1$. That is, 2.8R 1.9, 1.9R 1.1, but $2.8 \sim R$ 1.1.

Recurrence Relations

A *recurrence relation* for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence, namely, $a_0, a_1, \ldots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer. A sequence is called a *solution* of a recurrence relation if its terms satisfy the recurrence relation. (A recurrence relation is said to *recursively define* a sequence. \

Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 1, 2, 3, \ldots$, and suppose that $a_0 = 2$. What are a_1, a_2, and a_3?

Solution: We see from the recurrence relation that $a_1 = a_0 + 3 = 2 + 3 = 5$. It then follows that $a_2 = 5 + 3 = 8$ and $a_3 = 8 + 3 = 11$.

Recurrence Relations

Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} - a_{n-2}$ for $n = 2, 3, 4, \ldots$, and suppose that $a_0 = 3$ and $a_1 = 5$. What are a_2 and a_3?

Solution: We see from the recurrence relation that $a_2 = a_1 - a_0 = 5 - 3 = 2$ and $a_3 = a_2 - a_1 = 2 - 5 = -3$. We can find a_4, a_5, and each successive term in a similar way. ◀

The *Fibonacci sequence*, $f_0, f_1, f_2, \ldots$, is defined by the initial conditions $f_0 = 0$, $f_1 = 1$, and the recurrence relation

$$f_n = f_{n-1} + f_{n-2}$$

for $n = 2, 3, 4, \ldots$.

Recurrence Relations

Find the Fibonacci numbers f_2, f_3, f_4, f_5, and f_6.

Solution: The recurrence relation for the Fibonacci sequence tells us that we find successive terms by adding the previous two terms. Because the initial conditions tell us that $f_0 = 0$ and $f_1 = 1$, using the recurrence relation in the definition we find that

$$f_2 = f_1 + f_0 = 1 + 0 = 1,$$
$$f_3 = f_2 + f_1 = 1 + 1 = 2,$$
$$f_4 = f_3 + f_2 = 2 + 1 = 3,$$
$$f_5 = f_4 + f_3 = 3 + 2 = 5,$$
$$f_6 = f_5 + f_4 = 5 + 3 = 8.$$

Examples of Recurrence Relations

Suppose that $\{a_n\}$ is the sequence of integers defined by $a_n = n!$, the value of the factorial function at the integer n, where $n = 1, 2, 3, \ldots$. Because $n! = n((n-1)(n-2)\ldots 2 \cdot 1) = n(n-1)! = na_{n-1}$, we see that the sequence of factorials satisfies the recurrence relation $a_n = na_{n-1}$, together with the initial condition $a_1 = 1$.

We say that we have solved the recurrence relation together with the initial conditions when we find an explicit formula, called a **closed formula**, for the terms of the sequence.

Forward substitution – We find successive terms beginning with the initial condition and ending with a_n.

Backward substitution, We begin with a_n and iterated to express it in terms of falling terms of the sequence until we find it in terms of a_1.

Examples of Recurrence Relations

Determine whether the sequence $\{a_n\}$, where $a_n = 3n$ for every nonnegative integer n, is a solution of the recurrence relation $a_n = 2a_{n-1} - a_{n-2}$ for $n = 2, 3, 4, \ldots$. Answer the same question where $a_n = 2^n$ and where $a_n = 5$.

Solution: Suppose that $a_n = 3n$ for every nonnegative integer n. Then, for $n \geq 2$, we see that $2a_{n-1} - a_{n-2} = 2(3(n-1)) - 3(n-2) = 3n = a_n$. Therefore, $\{a_n\}$, where $a_n = 3n$, is a solution of the recurrence relation.

Suppose that $a_n = 2^n$ for every nonnegative integer n. Note that $a_0 = 1, a_1 = 2$, and $a_2 = 4$. Because $2a_1 - a_0 = 2 \cdot 2 - 1 = 3 \neq a_2$, we see that $\{a_n\}$, where $a_n = 2^n$, is not a solution of the recurrence relation.

Suppose that $a_n = 5$ for every nonnegative integer n. Then for $n \geq 2$, we see that $a_n = 2a_{n-1} - a_{n-2} = 2 \cdot 5 - 5 = 5 = a_n$. Therefore, $\{a_n\}$, where $a_n = 5$, is a solution of the recurrence relation.

Tutorial 3

1. Let A be the set of students who live within one mile of school and let B be the set of students who walk to classes. Describe the students in each of these sets.

a) $A \cap B$ **b)** $A \cup B$ **c)** $A - B$ **d)** $B - A$

2. Draw the Venn diagrams for each of these combinations of the sets A, B, C, and D.

a) $(A \cap B) \cup (C \cap D)$ **b)** $A \cup B \cup C \cup D$ **c)** $A - (B \cap C \cap D)$

3. Let A, B, and C be sets. Show that **a)** $(A \cup B) \subseteq (A \cup B \cup C)$ **b)** $(A \cap B \cap C) \subseteq (A \cap B)$.

4. The **symmetric difference** of A and B, denoted by $A \oplus B$, is the set containing those elements in either A or B, but not in both A and B. What can you say about the sets A and B if $A \oplus B = A$?

5. Show that if A, B, and C are sets, then $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$.

Tutorial 4

1. Why is f not a function from **R** to **R** if

a) $f(x) = 1/x$?

b) $f(x) = \sqrt{x}$?

c) $f(x) = \pm\sqrt{(x^2 + 1)}$?

2. Find the domain and range of these functions.

a) the function that assigns to each pair of positive integers the maximum of these two integers

b) the function that assigns to each positive integer the number of the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 that do not appear as decimal digits of the integer

3. Determine whether the function $f : \mathbf{Z} \times \mathbf{Z} \rightarrow \mathbf{Z}$ is onto if

a) $f(m, n) = m + n$.

b) $f(m, n) = m^2 + n^2$.

4. Determine whether each of these functions from {a, b, c, d} to itself is one-to-one.

a) f (a) = b, f (b) = a, f (c) = c, f (d) = d

b) f (a) = b, f (b) = b, f (c) = d, f (d) = c

c) f (a) = d, f (b) = b, f (c) = c, f (d) = d

5. Find f ◦ g and g ◦ f , where f (x) = x2 + 1 and g(x) = x + 2, are functions from R to R.

THANK YOU